

EDUCATION

University of California, Berkeley

M.S. in EECS, Distributed Systems for Machine Learning. Advised by James Demmel

Berkeley, CA

05/2025

University of California, Berkeley

Bachelors in Computer Science; Statistics; Mathematics (Triple Major) GPA: 3.9

05/2024

Organizations: CS Honor Society Officer, CSUA (CS Club Executive Officer), CSM (CS 61A/B/70/88 Tutor)

Relevant Coursework: Algos, Programming Languages & Compilers, Data Mining, Full Stack Web[†], LinAlg, Multivar Calc., Parallel Systems[†], Probability[†], Signal Processing, Robotics[†], Computer {Architecture, Graphics[†], Security, Vision[†]}, UI/UX, HCI[†], Neural Nets[†], Deep RL[†], NLP[†], Forecasting[†], {Real, Complex[†], Numerical[†]} Analysis[†], Theoretical Stats[†].

[†]= Graduate

EXPERIENCE

• Schonfeld (Quantitative Hedge Fund)

Applied Machine Learning Engineer Intern

New York, NY

Apr 2023 - Aug 2023

- Exploratory Data Analysis using **Databricks**' PySpark >100 GB of alternative data from AWS S3 and EC2, with storage optimized via Parquet files, for a statistically backtested signal. Under Systematic Trading team (QIS).
- Wrote an **Online** Algorithm to improve latency by >50% via Statistical Estimation for a **HFT Stat Arb** strategy. Measured effects of out-of-sample distributional shift via **divergence** measures. Wrote scheduler to ingest live data.

• Arrowstreet Capital LP

Software Developer Intern @ Arrowstreet's Research Core Team

Boston, MA

Jun 2022 - Aug 2022

- Fine-tuned a local **FinBERT** model from **Hugging Face** on over 100,000 vendor filings of various languages & correlated sentiments with word embeddings using **Cosine Similarity**, Jaccard Index, and Minkowski Distances.
- Wrote distributed code with **Numba** and **Dask** to discover hidden signals for equity alpha research. Made robust improvements to the firm-wide single systematic process with equities toolings via **Bayesian Regressions**.

• Electrical Engineering & Computer Sciences @ University of California, Berkeley

Researcher, Research Assistant;

Teaching: Instructor, Head Teaching Assistant, uGSI, Tutor, Reader Jan 2021 - Present

Berkeley, CA

- Lead TA - CS C100/200 (Data Science & Machine Learning), CS 160 (HCI), CS 164 (Compilers):** Managed admin logistics for ~300 students via the EdStem Q&A platform, assignment extensions (automated using Gradescope API), Incomplete Student assignment linkages + communications, held OH and taught graduate sections
- Head Teaching Assistant - EECS 122: Communication Networking/Internet:** Lead all administrative course logistics, **created** all existing course **content**, held office hours and **taught section** on-campus
- Reader, Academic Student Employee:** Graded (and wrote) exams on Data Structures (Lists/HashMaps/Heaps/Trees/etc) & Discrete Math (Logic, Crypto, Markov Chains, etc) & held Office Hours for **800+ students**
- Berkeley Research - Artificial Intelligence:** Deep Learning & **NLP:** Trained an **LSTM** to predict student grades from attendance and analyzed **Time Series** to determine earnestness with new SOTA **metrics** (Paper was accepted in ACM's 2023 conference: <https://arxiv.org/abs/2311.10746>). Also applied Deep RL techniques to 'solve' Minecraft. Peer reviewed abstract available: cmt3.research.microsoft.com/api/DEEPRLFP2022/Files/124

PROJECTS

- Live Streaming NeRF Nerfstudio Plugin:** Live re-rendering of shaders and textures in a game environment in real-time with minimal latency, with efficient C++ code, as well as adaptive noise reduction with anisotropic filters. Used a novel networking protocol to parse and transmit the live data from PyTorch, without loss or unmatched data pairs.
- GPU-accelerated Factorization Machines:** Boosted recommender systems training speed by **21.4%** using a novel randomized algorithm that capitalized on data **sparsity** to parallelize Stochastic Gradient Descent. Achieved via custom CUDA structures for efficient **low-rank** polynomial kernel approximation. Profiled bottlenecks with VTune's Roofline.
- Berkeley AI Research (BAIR) Lab:** Collaborated with Simeon Adebola (PhD candidate) and Professor Goldberg from AUTOLab to develop a robust **automated** cable insertion system for **dynamic** environments. Applied CV and flood-fill algorithms for cable & channel identification using perception, segmentation, tracing & endpoint **detection**.

AWARDS

- Donald M. Rizzo Memorial Scholarship Recipient:** Awarded from international NPO: Lions Club International
- Winner of the Congressional App Challenge:** Invited to Capitol Hill to demo mobile app to House of Representative members at #HouseOfCode, the most prestigious student STEM expo, at the U.S. Capitol Building
- Capture The Flag Security Competitions:** Ranked **top 50** in all of {PicoCTF, HSCTF, BCTF, CSAW Red}

SKILLS

- **Languages:** C++11, CUDA, Python, Typescript, GoLang, Java, Rust, SQL, Kotlin, Dart, R, Elixir, Lisp, C#
- **Technologies:** NeRFs, Flask, Android, Flutter, React Native, Node.js, Firebase, PyTorch, Pandas, Git, RegEx, L^AT_EX